EE3032 Introduction to VLSI Design—Final

1. (30 points) Answer the following questions.
   (a) Explain the latchup phenomenon and list two approaches to alleviate the occurrence probability of latchup. (6 points)
   (b) List two requirements for drawing the layout of a cell in the standard cell library. (4 points)
   (c) What is soft error? (2 points)
   (d) The soft error rate of a circuit is increased with the increasing of operation frequency and the scaling down of technology node? Explain the reasons in terms of three effects: logical masking effect, electrical masking effect, and latch window effect. (6 points)
   (e) List two limitations of the dynamic logic. (4 points)
   (f) What are the core-limited design and the pad-limited design? (4 points)
   (g) What is the physical design? What is the goal of the design rule? (4 points)

2. (10 points) Fig. 1 shows a domino logic realizing the Boolean function $Y = ABC$. Complete the following questions.

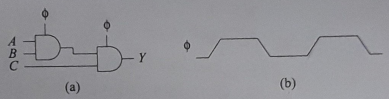


Figure 1: (a) A domino logic circuit (b) clock waveform

   (a) Draw the transistor-level schematic diagram of the domino logic. (3 points)
   (b) Assume that the t_r (rise time) and t_f (fall time) of each 2-input AND gate are $1ns$ and $2ns$, respectively. How long is the minimum clock period of the ϕ if the duty cycle of the ϕ is 50%? (4 points)
   (c) Assume that each transistor has the following characteristics: $C_d = 0.1fF$; $C_g = 1fF$; $I_{Lp} = 1nA$ (PMOS leakage current); $I_{Ln} = 2nA$ (NMOS leakage current). Also, the supply voltage=1.8V and the threshold voltage of logic 1 is 1.3V. Could the circuit be operated correctly at the minimum clock period? Why? (3 points)

3. (10 points) Figure 2 shows a Schmitt trigger circuit. If you apply an input signal as shown in the left figure to the Schmitt trigger circuit, the circuit outputs the signal as shown in the right figure. Assume that $\beta_n = 2\beta_p$; the turn-on equivalent resistance of N_1 transistor is R; the simple RC delay model is used to estimate the delay and $t_r = t_f = 2RC$. Complete the following questions.
   (a) Let $V_{tn} = |V_{tp}| = 0.4V$, $V_{DD} = 2.0V$, $V_T^+ = 1.4V$, and $V_T^- = 0.6V$. If the size of $N_1 = 100nm/40nm$, determine the sizes of the transistors N_2, N_3, P_1, P_2, and P_3. (5 points)
   (b) Let $V_{tn} = |V_{tp}| = 0.6V$, $V_{DD} = 1.8V$, $V_T^+ = 1.2V$, and $V_T^- = 0.6V$. If an inverter is designed by removing the transistors N_3 and P_3, is the short-circuit power of the inverter larger or smaller than that of the Schmitt trigger circuit? Explain your answer. (5 points)

4. (15 points) Draw the transistor-level schematic diagram of the Boolean function $f = \overline{A(BC + D)}$ using each of the following logic design styles: (1) Dynamic logic (5 points) (2) Fully differential logic (5 points) (3) Differential split-level logic (5 points).

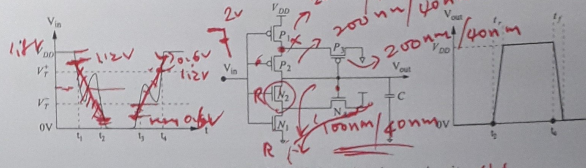


Figure 2: A CMOS Schmitt trigger circuit

5. (15 points) Figure 3 shows an NP-domino logic circuit. Assume that $\beta_n = 2\beta_p$. Complete the following questions.

   (a) What is the Boolean function of the NP-domino logic circuit? (5 points)

   (b) If the sizes of P_1 and N_5 are $80nm/20nm$ and $20nm/20nm$, respectively, determine the sizes of the other transistors such that the $t_r = t_f$ at nodes X and Y. (5 points)

   (c) If the gate capacitance of the N_5 is C, the supply voltage is V_{dd}, and the operation frequency is f, express the dynamic power consumed by the clock in terms of V_{dd}, f, and C. (5 points)

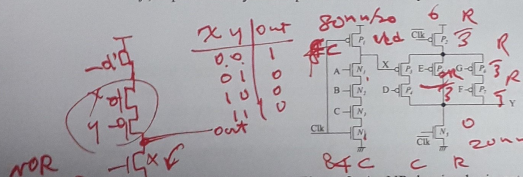


Figure 3: An NP-domino logic gate

6. (10 points) Draw the transistor-level schematic diagram of the symbolic layout shown in Figure 4.

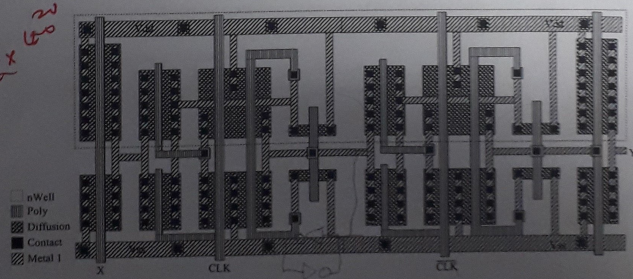


Figure 4: Symbolic layout

7. (10 points) Figure 5 shows one possible implementation of the D register. Complete the following questions.

(a) Figure 5 shows a register consisting of a master latch and a low-cost slave latch. Label the clock signals for the transmission gates such that the register is a negative-edge triggered register. (3 points)

(b) Assume that the delay of Inv1, Inv2, and transmission gate is 0.15ns, 0.2ns, and 0.1ns, respectively. Define the value of setup time of the register if the clock is ideal. (2 points)

(c) As Figure 5 shows, assume that the switching threshold voltage of the inverter Inv2 is $V_{dd}/2$ and the turn-on equivalent resistances of the transmission gate, N1, N2, P1, and P2 are R_{TS}, R_{N1}, R_{N2}, R_{P1}, and R_{P2}, respectively. List two inequalities for determining the sizes of P2 and N2 such that the register can be operated correctly. (5 points)

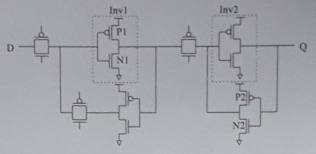


Figure 5: A static register

0.2ns

8. (15 points) Figure 6 shows a sequential circuit. Assume that the clock-to-Q propagation delay (t_{pcq}), the clock-to-Q contamination delay (t_{ccq}), the setup time (t_{setup}), and the hold time (t_{hold}) of the registers are $0.2ns$, $0.1ns$, $0.2ns$, and $0.2ns$, respectively. The delay of the XNOR gate, the AND gate, the OR gate, and the inverter is $0.4ns$, $0.5ns$, $0.5ns$, and $0.2ns$, respectively. Also, the delay of interconnection can be ingnored. Complete the following questions.

(a) Can the circuit be operated correctly? If not, list two possible methods to modify the circuit such that it can operate correctly without sacrificing the operation frequency? Draw the modified circuit. (5 points) ① buffer insertion ② lockup latch

(b) Determine the maximum operation frequency of the modified circuit. (5 points)

(c) Inserting registers into the circuit such that the modified circuit can be operated at $\frac{4}{3}$ times of the original operation frequency. (5 points)

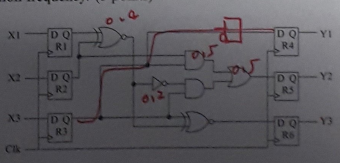


Figure 6: A sequential circuit

$T_c \geq t_{pcq} + t_{pd} + t_{setup}$

$0.2 + (0.4 + 0.2 + 0.5) + 0.2$ 2ns